

Serverless Architecture & Application Flow

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Serverless architecture is a modern cloud-computing model where developers focus completely on writing application logic while the cloud provider (AWS) handles all infrastructure responsibilities such as server provisioning, scaling, patching, and availability. Instead of running applications on traditional servers, your code executes only when required, leading to reduced operational overhead, auto-scaling, and cost savings because you pay only for the actual usage.

In this project, we are using a fully managed AWS serverless stack consisting of **Amazon S3, API Gateway, AWS Lambda, DynamoDB, and Amazon Cognito**. Each service plays a unique role and together forms a robust, secure, and scalable architecture for web applications.

Amazon S3 acts as the hosting platform for the frontend of the application. It stores and delivers static files such as HTML, CSS, JavaScript, and images. Since S3 is highly durable and optimized for global content delivery, users experience fast loading times without needing a traditional web server.

Amazon API Gateway is the communication bridge between the frontend and backend. Whenever the user interacts with the web interface (e.g., login, submit form, fetch data), the browser sends a request to API Gateway. API Gateway then handles request validation, throttling, authentication checks, and securely forwards the request to the appropriate backend Lambda function.

AWS Lambda contains the entire backend logic of the system. It runs code in response to events without requiring any server setup. Each function is triggered only when called by API Gateway, which significantly reduces cost and ensures automatic scaling during high traffic. Lambda interacts directly with DynamoDB to fetch, store, or update application data.

Amazon DynamoDB is a fast, fully managed NoSQL database that stores user records, application state, and other data required by the system. It seamlessly integrates with Lambda, provides single-digit millisecond performance, and automatically scales to handle any amount of traffic. Its serverless nature makes it ideal for modern cloud applications.

Amazon Cognito manages the entire user authentication and authorization process. When users sign up or log in, Cognito verifies their identity and generates secure JSON Web Tokens (JWT). These tokens are then validated by API Gateway before allowing any user to access protected resources or APIs. This ensures that only authenticated and authorized users can interact with the backend.

Overall Application Flow

1. The user opens the website, and the static content is delivered from Amazon S3.
2. When the user performs an action, the frontend sends a request to Amazon API Gateway.
3. API Gateway verifies authentication (via Cognito tokens) and routes the request to Lambda.
4. The Lambda function executes the required business logic.
5. Lambda interacts with DynamoDB to read or write application data.

6. API Gateway sends the processed response back to the frontend for the user to view.

This architecture is highly scalable, secure, and cost-efficient. Since all components are fully managed, developers can concentrate on building features rather than maintaining servers. The combination of these AWS services creates a seamless, elastic, and modern application environment suitable for production-grade web applications.

ARCHITECHTURE FLOW:

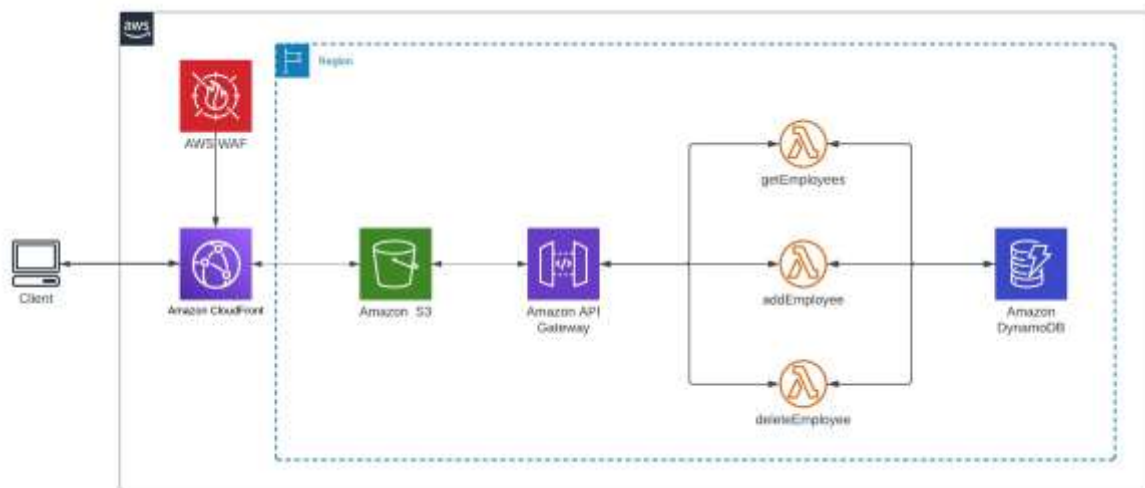


FIG.01 – Serverless Architecture Overview

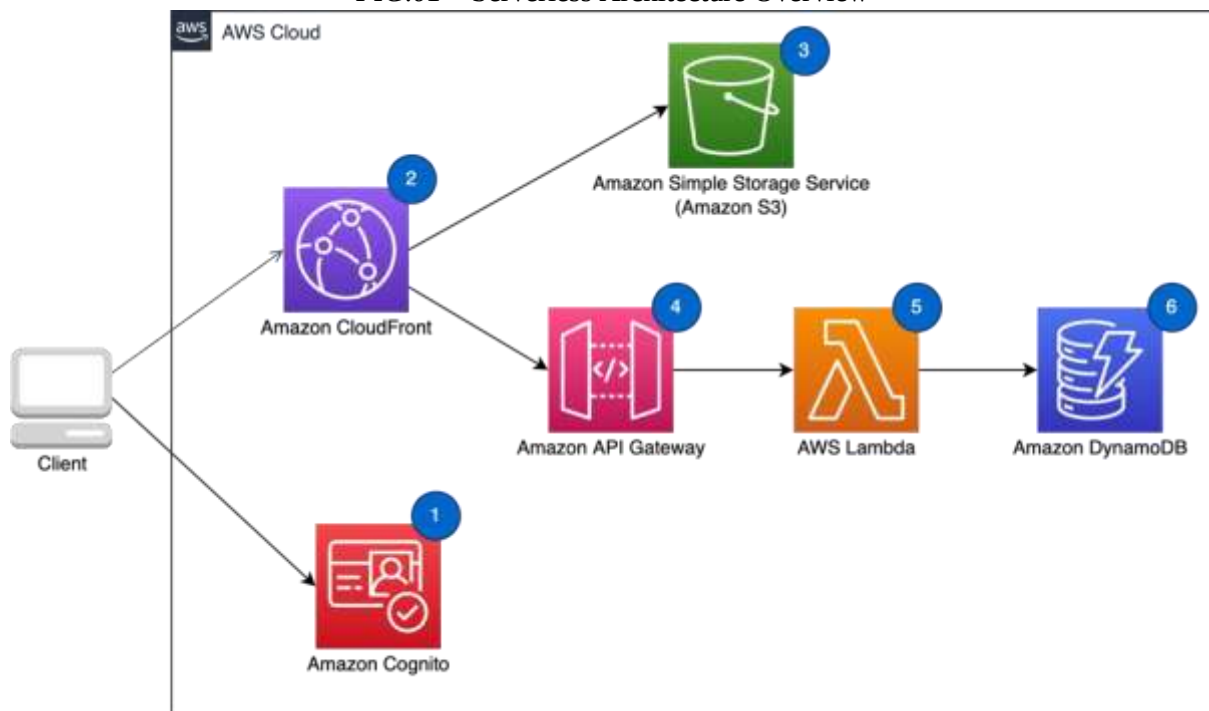


FIG.02 – Application Flow Diagram